

ADA PROJECT

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Programme: BTECH CSE AI & ML

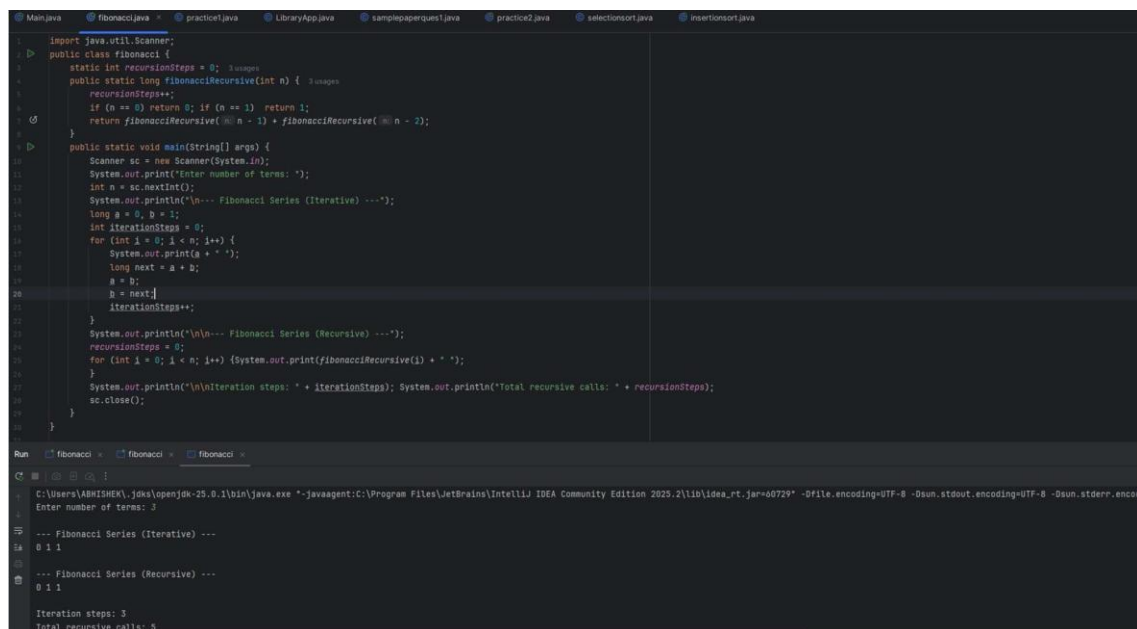
Section: C

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Question 1: Fibonacci's Series



```
1 import java.util.Scanner;
2 public class fibonacci {
3     static int recursionSteps = 0;
4     public static long fibonacciRecursive(int n) {
5         recursionSteps++;
6         if (n == 0) return 0; if (n == 1) return 1;
7         return fibonacciRecursive(n - 1) + fibonacciRecursive(n - 2);
8     }
9     public static void main(String[] args) {
10         Scanner sc = new Scanner(System.in);
11         System.out.print("Enter number of terms: ");
12         int n = sc.nextInt();
13         System.out.println("\n--- Fibonacci Series (Iterative) ---");
14         long a = 0, b = 1;
15         int iterationSteps = 0;
16         for (int i = 0; i < n; i++) {
17             System.out.print(a + " ");
18             long next = a + b;
19             a = b;
20             b = next;
21             iterationSteps++;
22         }
23         System.out.println("\n--- Fibonacci Series (Recursive) ---");
24         recursionSteps = 0;
25         for (int i = 0; i < n; i++) {System.out.print(fibonacciRecursive(i) + " ");
26         }
27         System.out.println("\nIteration steps: " + iterationSteps); System.out.println("Total recursive calls: " + recursionSteps);
28         sc.close();
29     }
30 }
```

Run

```
C:\Users\ABHISHEK\jksa\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2023.2\lib\idea_rt.jar=60729" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Enter number of terms: 3

--- Fibonacci Series (Iterative) ---
0 1 1

--- Fibonacci Series (Recursive) ---
0 1 1

Iteration steps: 3
Total recursive calls: 5
```

Question 2: Factorial Number:

```
public class Factorial {
    static int stepCount = 0; //usage
    static void factIterative(int n) { //usage
        stepCount = 0;
        int factorial = 1;
        for (int i = 1; i <= n; i++) {
            factorial = factorial * i;
            stepCount++;
        }
        System.out.println("Iterative Factorial: " + factorial);
        System.out.println("Step Count: " + stepCount);
    }
    static int factRecursive(int n) { //usage
        stepCount++;
        if (n <= 1)
            return 1;
        return n * factRecursive(n - 1);
    }
    public static void main(String[] args) {
        int n = 5;
        factIterative(n);
        stepCount = 0;
        int result = factRecursive(n);
        System.out.println("Recursive Factorial: " + result);
        System.out.println("Step Count: " + stepCount);
    }
}
```

Run

C:\Users\ABHISHEK\.jdk\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=54451 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8

Iterative Factorial: 120
Step Count: 5
Recursive Factorial: 120
Step Count: 5
Process finished with exit code 0

Question 3: Insertion Sort

```
import java.util.Scanner;

public class InsertionSort {
    public static void insertionSort(int[] arr) { //usage
        int n = arr.length; int comparisons = 0; int shifts = 0; int outerIterations = 0;
        for (int i = 1; i < n; i++) {
            outerIterations++; int key = arr[i]; int j = i - 1;
            while (j >= 0 & (arr[j] > key)) { arr[j + 1] = arr[j]; shifts++; j--; }
            arr[j + 1] = key;
        }
        System.out.println("\nSorted Array:");
        for (int num : arr) {
            System.out.print(num + " ");
        }
        System.out.println("\nOuter Iterations: " + outerIterations);
        System.out.println("Total Comparisons: " + comparisons);
        System.out.println("Total Shifts: " + shifts);
    }
    public static void main(String[] args) { Scanner sc = new Scanner(System.in); System.out.print("Enter number of elements: ");
        int n = sc.nextInt(); int[] arr = new int[n]; System.out.println("Enter elements:");
        for (int i = 0; i < n; i++) { arr[i] = sc.nextInt(); }
        insertionSort(arr); sc.close();
    }
}
```

Run

C:\Users\ABHISHEK\.jdk\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=62114 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8

Enter number of elements: 3
Enter elements:
1
2
3
Sorted Array:
1 2 3
Outer Iterations: 2
Total Comparisons: 2
Total Shifts: 0
Process finished with exit code 0

Question 4: Merge Sort

```

Main.java  fibonacci.java  Factorial.java  practice1.java  LibraryApp.java  samplepaperquest.java  practice2.java  selectionsort.java  insertionsort.java  MergeSort.java
public class MergeSort {
    static int stepCount = 0; static void mergeSort(int[] arr, int left, int right) {
        if (left < right) { int mid = (left + right) / 2; mergeSort(arr, left, mid); mergeSort(arr, mid + 1, right); merge(arr, left, mid, right); }
        static void merge(int[] arr, int left, int mid, int right) { int n1 = mid - left + 1; int n2 = right - mid; int[] L = new int[n1]; int[] R = new int[n2];
            for (int i = 0; i < n1; i++) L[i] = arr[left + i];
            for (int j = 0; j < n2; j++) R[j] = arr[mid + 1 + j];
            int i = 0, j = 0, k = left;
            while (i < n1 && j < n2) { stepCount++; // comparison
                if (L[i] <= R[j]) { arr[k++] = L[i++]; }
                else { arr[k++] = R[j++]; }
            }
            while (i < n1) {
                arr[k++] = L[i++];
            }
            while (j < n2) {
                arr[k++] = R[j++];
            }
        }
        static void printArray(int[] arr) {
            for (int num : arr)
                System.out.print(num + " ");
            System.out.println();
        }
        public static void main(String[] args) {
            int[] arr = {35, 12, 48, 7, 22};
            stepCount = 0;
            mergeSort(arr, 0, arr.length - 1);
            System.out.println("Sorted Array (Merge Sort):");
            printArray(arr);
            System.out.println("Step Count: " + stepCount);
        }
    }
}
Run MergeSort
C:\Users\ABHISHEK\jdk\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=50149* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Sorted Array (Merge Sort):
7 12 22 35 48
Step Count: 7
Process finished with exit code 0

```

Question 5: Quick Sort

```

Main.java  fibonacci.java  Factorial.java  practice1.java  LibraryApp.java  samplepaperquest.java  practice2.java  selectionsort.java  insertionsort.java  MergeSort.java  QuickSort.java
public class QuickSort {
    static int stepCount = 0; static void quickSort(int[] arr, int low, int high) {
        if (low < high) { int pi = partition(arr, low, high); quickSort(arr, low, pi - 1); quickSort(arr, pi + 1, high); }
    }
    static int partition(int[] arr, int low, int high) {
        int pivot = arr[high]; int i = low - 1;
        for (int j = low; j < high; j++) { stepCount++; // comparison
            if (arr[j] < pivot) { i++; swap(arr, j, i); }
        }
        swap(arr, i + 1, high); return i + 1;
    }
    static void swap(int[] arr, int i, int j) {
        int temp = arr[i];
        arr[i] = arr[j];
        arr[j] = temp;
        stepCount++; // swap count
    }
    static void printArray(int[] arr) {
        for (int num : arr)
            System.out.print(num + " ");
        System.out.println();
    }
    public static void main(String[] args) {
        int[] arr = {35, 12, 48, 7, 22};
        stepCount = 0;
        quickSort(arr, 0, arr.length - 1);
        System.out.println("Sorted Array (Quick Sort):");
        printArray(arr);
        System.out.println("Step Count: " + stepCount);
    }
}
Run QuickSort
C:\Users\ABHISHEK\jdk\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=59967* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Sorted Array (Quick Sort):
7 12 22 35 48
Step Count: 12
Process finished with exit code 0

```

Question 6: Selection Sort

```
1 import java.util.Scanner;
2 public class selectionSort {
3     public static void selectionSort(int[] arr) {
4         int n = arr.length; int comparisonSteps = 0; int swapSteps = 0;
5         for (int i = 0; i < n - 1; i++) { int minIndex = i;
6             for (int j = i + 1; j < n; j++) { comparisonSteps++;
7                 if (arr[j] < arr[minIndex]) { minIndex = j;
8                     }
9                 }
10            if (minIndex != i) { int temp = arr[i]; arr[i] = arr[minIndex]; arr[minIndex] = temp; swapSteps++;
11            }
12        }
13        System.out.println("Unsorted Array:");
14        for (int num : arr) {
15            System.out.print(num + " ");
16        }
17        System.out.println("\nTotal Comparisons: " + comparisonSteps); System.out.println("Total Swaps: " + swapSteps);
18    }
19    public static void main(String[] args) {
20        Scanner sc = new Scanner(System.in); System.out.print("Enter number of elements: ");
21        int n = sc.nextInt(); int[] arr = new int[n]; System.out.println("Enter elements:");
22        for (int i = 0; i < n; i++) {
23            arr[i] = sc.nextInt();
24        }
25        selectionSort(arr); sc.close();
26    }
27 }
```

Run selectionSort

C:\Users\ABHISHEK\jdk\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=49420 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8

Enter number of elements: 3

Enter elements:

12

13

14

Sorted Array:

12 13 14

Total Comparisons: 3

Total Swaps: 0

Process finished with exit code 0

Question 7: Bubble Sort

```
1 public class BubbleSort {
2     public static void main(String[] args) {
3         int[] arr = {9, 5, 1, 4, 3};
4         bubbleSort(arr);
5         for (int num : arr) {
6             System.out.print(num + " ");
7         }
8     }
9     public static void bubbleSort(int[] arr) {
10        int n = arr.length;
11        boolean swapped;
12        for (int i = 0; i < n - 1; i++) {
13            swapped = false;
14            for (int j = 0; j < n - i - 1; j++) {
15                if (arr[j] > arr[j + 1]) {
16                    int temp = arr[j];
17                    arr[j] = arr[j + 1];
18                    arr[j + 1] = temp;
19                    swapped = true;
20                }
21            }
22            if (!swapped) break;
23        }
24    }
25 }
```

Run BubbleSort

C:\Users\ABHISHEK\jdk\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=57423 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8

1 3 4 5 9

Process finished with exit code 0

Question 8: Linear Search

```
1 import java.util.concurrent.ThreadLocalRandom;
2 import java.util.Scanner;
3
4 public class LinearSearch {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         int[] arr = new int[5];
8         System.out.println("Enter array element:");
9         for(int i=0; i<arr.length; i++) {
10             arr[i] = sc.nextInt();
11         }
12         System.out.println("\nArray elements:");
13         for(int i=0; i<arr.length; i++) {
14             System.out.print(i + " ");
15         }
16         System.out.println("\nLinear Search:");
17         linearSearch(arr, 20);
18     }
19
20     public static void linearSearch(int[] arr, int target) { //usage
21         boolean isFound = false;
22         for(int i=0; i<arr.length; i++) {
23             if(arr[i] == target) {
24                 System.out.println("element found on: " + arr[i]);
25                 isFound = true;
26                 break;
27             }
28         }
29         if(!isFound) System.out.println("element not found in list!");
30     }
31 }
32
33 Run LinearSearch
34
35 C:\Users\ABHISHEK\.jws\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=55250* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
36
37 Enter array element:
38 2
39 3
40 4
41
42 Array elements
43 0 1 2
44 Linear Search
45 element not found in list!
46
47 Process finished with exit code 0
```

Question 9: Binary Search

```
1 public class BinarySearch {
2     static int stepCount = 0; //usage
3     static int binarySearchIterative(int[] arr, int key) { stepCount = 0; int low = 0; int high = arr.length - 1; //usage
4         while (low <= high) { stepCount++; int mid = (low + high) / 2;
5             if (arr[mid] == key) return mid;
6             else if (arr[mid] < key) low = mid + 1;
7             else high = mid - 1;
8         } return -1;
9     }
10
11     static int binarySearchRecursive(int[] arr, int low, int high, int key) { stepCount++; //usage
12         if (low > high) return -1; int mid = (low + high) / 2;
13         if (arr[mid] == key) return mid;
14         else if (arr[mid] < key) return binarySearchRecursive(arr, mid + 1, high, key);
15         else return binarySearchRecursive(arr, low, mid - 1, key);
16     }
17
18     public static void main(String[] args) {int[] arr = {9, 12, 22, 35, 48}; int key = 22; int result1 = binarySearchIterative(arr, key);
19         if (result1 != -1) System.out.println("Iterative: Element found at index " + result1);
20         else System.out.println("Iterative: Element not found");
21         System.out.println("Step Count (Iterative): " + stepCount);
22         stepCount = 0;
23         int result2 = binarySearchRecursive(arr, 0, arr.length - 1, key);
24         if (result2 != -1)
25             System.out.println("Recursive: Element found at index " + result2);
26         else System.out.println("Recursive: Element not found"); System.out.println("Step Count (Recursive): " + stepCount);
27     }
28 }
29
30 Run BinarySearch
31
32 C:\Users\ABHISHEK\.jws\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=58602* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
33
34 Iterative: Element found at index 2
35 Step Count (Iterative): 1
36 Recursive: Element found at index 2
37 Step Count (Recursive): 1
38
39 Process finished with exit code 0
```

Question 10 : Bellman-Ford

```
1 import java.util.Scanner;
2 class Edge { int u, v, w; Edge(int u, int v, int w) { this.u = u; this.v = v; this.w = w; } }
3 public class Bellman_Ford { static void bellmanFord(Edge[] edges, int V, int E, int src) { int[] dist = new int[V];
4     for (int i = 0; i < V; i++) { dist[i] = Integer.MAX_VALUE;
5     } dist[src] = 0;
6     for (int i = 1; i <= V - 1; i++) { for (int j = 0; j < E; j++) { int u = edges[j].u; int v = edges[j].v; int w = edges[j].w;
7         if (dist[u] != Integer.MAX_VALUE && dist[u] + w < dist[v]) { dist[v] = dist[u] + w; } } }
8     for (int j = 0; j < E; j++) { int u = edges[j].u; int v = edges[j].v; int w = edges[j].w;
9     if (dist[u] != Integer.MAX_VALUE && dist[u] + w < dist[v]) { System.out.println("negative weight cycle detected"); return; } }
10    System.out.println("Vertex\tDistance"); for (int i = 0; i < V; i++) { System.out.print(i + "\t" + dist[i]); } }
11    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int V, E;
12    do { System.out.print("Enter number of vertices (<4): "); V = sc.nextInt();
13    } while (V <= 0 || V >= 4); do { System.out.print("Enter number of edges (<4): "); E = sc.nextInt();
14    } while (E <= 0 || E >= 4); Edge[] edges = new Edge[E]; System.out.println("Enter edges (u v w):");
15    for (int i = 0; i < E; i++) { int u = sc.nextInt(); int v = sc.nextInt(); int w = sc.nextInt();
16    if (u >= V || v >= V || u < 0 || v < 0) { System.out.println("Invalid vertex: Re-enter edge"); i--; continue;
17    } edges[i] = new Edge(u, v, w); }
18    int src; do {
19        System.out.print("Enter source vertex: ");
20        src = sc.nextInt();
21    } while (src < 0 || src >= V);
22    bellmanFord(edges, V, E, src);
23    sc.close();
24    }
25    }
```

Run Bellman_Ford

```
C:\Users\ADMIN\jks\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=52550 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Enter number of vertices (<4): 3
Enter number of edges (<4): 3
Enter edges (u v w):
0 1 4
1 2 -2
0 2 5
Enter source vertex: 0
Vertex Distance
0 0
1 4
2 2
Process finished with exit code 0
```

Question 11: Dijkstra Algorithm

```
1 import java.util.Scanner;
2 public class DijkstraAlgorithm {
3     static int minDistance(int[] dist, boolean[] visited, int V) { int min = Integer.MAX_VALUE, min_index = -1;
4     for (int v = 0; v < V; v++) { if (!visited[v] && dist[v] <= min) { min = dist[v]; min_index = v; } }
5     return min_index;
6 }
7 static void bellmanFord(int[] graph, int V, int src) { int[] dist = new int[V]; boolean[] visited = new boolean[V];
8     for (int i = 0; i < V; i++) { dist[i] = Integer.MAX_VALUE; visited[i] = false;
9     } dist[src] = 0;
10    for (int count = 0; count < V - 1; count++) { int u = minDistance(dist, visited, V); if (u == -1) break; visited[u] = true;
11    for (int v = 0; v < V; v++) { if (!visited[v] && graph[u][v] != 0 && dist[u] != Integer.MAX_VALUE && dist[u] + graph[u][v] < dist[v]) { dist[v] = dist[u] + graph[u][v]; } }
12    System.out.println("Vertex\tDistance");
13    for (int i = 0; i < V; i++) { System.out.print(i + "\t" + dist[i]); } }
14    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int V;
15    do { System.out.print("Enter number of vertices (<4): "); V = sc.nextInt();
16    } while (V <= 0 || V >= 4); int[][] graph = new int[V][V]; System.out.println("Enter adjacency matrix:");
17    for (int i = 0; i < V; i++) { for (int j = 0; j < V; j++) { graph[i][j] = sc.nextInt(); } } int src;
18    do { System.out.print("Enter source vertex: "); src = sc.nextInt();
19    } while (src < 0 || src >= V); bellmanFord(graph, V, src);
20    sc.close();
21    }
22    }
```

Run DijkstraAlgorithm

```
C:\Users\ADMIN\jks\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=52550 -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Enter number of vertices (<4): 3
Enter adjacency matrix:
0 1 4
1 0 2
4 2 0
Enter source vertex: 0
Vertex Distance
0 0
1 1
2 3
Process finished with exit code 0
```

Question 12: Fractional Knapsack

```
import java.util.*;

class Item { int value, weight; Item(int value, int weight) {this.value = value; this.weight = weight;} }

public class FractionalKnapsack { static double fractionalKnapsack(int W, Item[] arr, int n) { Arrays.sort(arr, (Item a, Item b) -> { double r1 = (double) a.value / a.weight; double r2 = (double) b.value / b.weight; return Double.compare(r2, r1); }); double totalValue = 0.0; for (int i = 0; i < n; i++) { if (W >= arr[i].weight) { W -= arr[i].weight; totalValue += arr[i].value; } else { totalValue += arr[i].value * ((double) W / arr[i].weight); break; } } return totalValue; }

public static void main(String[] args) { Scanner sc = new Scanner(System.in); int n, W; do { System.out.print("Enter number of items (<4): "); n = sc.nextInt(); } while (n <= 0 || n >= 4); Item[] arr = new Item[n]; System.out.println("Enter value and weight of each item:"); for (int i = 0; i < n; i++) { int value = sc.nextInt(); int weight = sc.nextInt(); if (weight <= 0) { System.out.println("Invalid weight! Re-enter:"); i--; continue; } arr[i] = new Item(value, weight); } System.out.print("Enter capacity of knapsack: "); W = sc.nextInt(); double result = fractionalKnapsack(W, arr, n); System.out.println("Maximum value = " + result); sc.close(); } }

Run FractionalKnapsack
C:\Users\ABHISHEK\.jdev\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=59459* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Enter number of items (<4): 3
Enter value and weight of each item:
40 10
100 20
120 30
Enter capacity of knapsack: 50
Maximum value = 240.0

Process finished with exit code 0
```

Question 13: Matrix Chain Multiplication

```
import java.util.Scanner;

public class MatrixChainMultiplication {
    static int matrixChainMultiplication(int[] p, int n) { int[][] dp = new int[100][100]; for (int i = 1; i < n; i++) { dp[i][i] = 0; } for (int L = 2; L <= n; L++) { for (int i = 1; i <= n - L + 1; i++) { int j = i + L - 1; dp[i][j] = Integer.MAX_VALUE; for (int k = i; k < j; k++) { int cost = dp[i][k] + dp[k+1][j] + p[i-1] * p[k] * p[j]; if (cost < dp[i][j]) { dp[i][j] = cost; } } } } return dp[1][n-1]; }

    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int n; do { System.out.print("Enter number of matrices (<4): "); n = sc.nextInt(); } while (n <= 0 || n >= 4); int[] p = new int[n+1]; System.out.println("Enter dimensions array (size " + (n+1) + "):"); for (int i = 0; i <= n; i++) { p[i] = sc.nextInt(); } int result = matrixChainMultiplication(p, n+1); System.out.println("Minimum multiplications = " + result); sc.close(); } }

Run MatrixChainMultiplication
C:\Users\ABHISHEK\.jdev\openjdk-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=61139* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8
Enter number of matrices (<4): 3
Enter dimensions array (size 4):
10 20 30 40
Minimum multiplications = 18000

Process finished with exit code 0
```

Question 14: Largest Common Subsequence

```
import java.util.Scanner;

public class LCS {
    static int lcs(String X, String Y, int m, int n) { int[][] dp = new int[100][100]; for (int i = 0; i <= m; i++) { for (int j = 0; j <= n; j++) { if (i == 0 || j == 0) dp[i][j] = 0; } }
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            if (X.charAt(i - 1) == Y.charAt(j - 1))
                dp[i][j] = 1 + dp[i - 1][j - 1];
            else
                dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
        }
    }
    return dp[m][n];
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    String X, Y;
    do {
        System.out.print("Enter first string (length < 4): ");
        X = sc.next();
    } while (X.length() >= 4);
    do {
        System.out.print("Enter second string (length < 4): ");
        Y = sc.next();
    } while (Y.length() >= 4);
    int m = X.length();
    int n = Y.length();
    System.out.println("Length of LCS = " + lcs(X, Y, m, n));
    sc.close();
}
}
```

Run LCS

C:\Users\ABHISEKH\jdoslogonjdw-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=52365* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8

Enter first string (length < 4): ab
Enter second string (length < 4): ac
Length of LCS = 1

Process finished with exit code 0

Question 15: 0/1 Fractional Knapsack

```
import java.util.Scanner;

public class ZeroesByOneKnapsack {
    static int knapsack(int W, int[] wt, int[] val, int n) { int[][] dp = new int[100][100];
    for (int i = 0; i <= n; i++) {
        for (int w = 0; w <= W; w++) { if (i == 0 || w == 0) dp[i][w] = 0;
        else if (wt[i - 1] <= w) dp[i][w] = Math.max(val[i - 1] + dp[i - 1][w - wt[i - 1]], dp[i - 1][w]);
        else dp[i][w] = dp[i - 1][w]; } return dp[n][W];
    }

    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int n, W;
    do { System.out.print("Enter number of items (<4): "); n = sc.nextInt();
    } while (n <= 0 || n >= 4); int[] val = new int[n]; int[] wt = new int[n];
    System.out.println("Enter values of items:");
    for (int i = 0; i < n; i++) {
        val[i] = sc.nextInt();
    }
    System.out.println("Enter weights of items:");
    for (int i = 0; i < n; i++) {
        wt[i] = sc.nextInt();
    }
    System.out.print("Enter capacity of knapsack: ");
    W = sc.nextInt();
    int result = knapsack(W, wt, val, n);
    System.out.println("Maximum value = " + result);
    sc.close();
}
}
```

Run ZeroesByOneKnapsack

C:\Users\ABHISEKH\jdoslogonjdw-25.0.1\bin\java.exe -javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=42155* -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding=UTF-8

Enter number of items (<4): 5
Enter values of items:
40 100 120
Enter weights of items:
10 20 30
Enter capacity of knapsack: 50
Maximum value = 220

Process finished with exit code 0